

نظام التشغيل العقاري المؤسسي المتكامل

REAL ESTATE OPERATING SYSTEM (REOS)

Enterprise-Grade Architecture & Development Blueprint

An all-inclusive, multi-tenant, AI-powered Enterprise Real Estate Operating System (REOS) covering the entire real estate life cycle — from lead capture, full sales CRM pipeline, integrated call center, marketing automation, dynamic pricing, and collections, to digital contracts, installments rescheduling, cancellations, advanced handover inspection, warranty management, vendor control, configurable workflow engine, OCR-based DMS, and executive BI analytics. Serving 100,000+ users across six deployment interfaces.

PROJECT NAME

REOS — Real Estate Operating
System

TARGET USERS

100,000+ Multi-Role Users

CLASSIFICATION

Confidential — v2.0 Enterprise
Expansion (2026)

? A. EXECUTIVE SUMMARY

الملخص التنفيذي

This document presents a comprehensive system analysis and digital platform blueprint for a large-scale real estate development company. The platform is designed as an all-inclusive Enterprise Real Estate Operating System (REOS) to digitally transform end-to-end operations — from lead generation, dynamic marketing tracking, CRM sales pipelines, unified call centers, unit reservations, through payment collection management, rescheduling, cancellations, handover inspections, warranty management, contractor SLAs, to post-handover community services and executive BI forecasting.

The proposed platform will serve 100,000+ users across multiple roles: direct clients, brokers, brokerage firms, sales teams, call center agents, collections agents, marketing managers, warranty technicians, vendor coordinators, finance teams, legal teams, project managers, and executive management. It will be deployed across six interfaces: iOS and Android mobile apps, a web/desktop application, an administration dashboard, a broker portal, and a client portal.

Powered by predictive AI, dynamic pricing models, fortified with enterprise-grade security, visual workflow builders, and OCR-driven Document Management Systems, this platform represents a strategic investment in operational efficiency, client experience, and sustainable competitive advantage.

ITEM	DETAILS
Project Name	Real Estate Operating System (REOS)
Target Users	100,000+ (Buyers, Brokers, Staff, Operations, Management)
Deployment	iOS Android Web Desktop Admin Dashboard Broker Portal Client Portal
Document Type	Business Analysis, System Design, Product Requirements, Architecture Blueprint
Version	2.0 — Enterprise Expansion Edition
Classification	Confidential — Authorized Enterprise Distribution Only

✓ B. PROJECT OBJECTIVE

هدف المشروع

The primary objective is to build a unified, secure, intelligent digital ecosystem that:

- Centralizes all real estate operations — sales, finance, contracts, maintenance — on one platform

- Provides each user type with a tailored experience aligned to their role and permissions
- Automates critical workflows including payment reminders, reservation queues, contract scheduling, and maintenance routing
- Integrates AI capabilities to reduce human error, detect fraud, and generate actionable insights
- Enforces compliance, audit trails, and data protection across all operations
- Enables brokers and brokerage firms to operate within a governed, transparent commission management framework
- Delivers a world-class post-handover experience including maintenance requests, community services, and utility management

! C. BUSINESS PROBLEM

المشكلة التجارية

Large real estate developers operating without a unified digital platform face the following critical challenges:

1 Fragmented Operations

- Sales, contracts, payments, and maintenance handled across multiple disconnected tools
- Lack of real-time visibility into unit availability, payment status, and project progress
- Manual processes prone to human error and data inconsistency

2 Broker Management Challenges

- No standardized system for broker registration, verification, or contract management
- Frequent disputes over client ownership, commission entitlement, and double registrations
- Inability to track broker performance, sales volume, or commission status in real time

3 Client Experience Gaps

- Clients lack transparency into their contract status, payment schedule, or delivery timelines
- No self-service portal for clients to manage payments, raise maintenance requests, or track unit status
- Inefficient new project launches create chaos in reservation queuing and client scheduling

4 Fraud and Compliance Risks

- Double reservations, unauthorized unit allocations, and impersonation risks
- No biometric identity verification or KYC-equivalent process
- Absence of audit trails makes internal misconduct difficult to detect or prosecute

5

Post-Sale Service Deficiency

- No structured system for maintenance requests, prioritization, or resolution tracking
- Residents have no digital channel for community services, visitor access, or utility requests

D. PROPOSED SOLUTION

الحل المقترح

The Real Estate Digital Platform (REDP) is a multi-tenant, role-based, AI-powered SaaS platform delivered across five interfaces:

PLATFORM	INTERFACE	PRIMARY USERS
Mobile App	iOS & Android	Clients, Brokers, Sales Staff, Technicians
Web Application	Browser-based Desktop	All Users — Extended Functionality
Admin Dashboard	Web — Restricted Access	Management, Finance, HR, Operations
Broker Portal	Web + Mobile View	Brokerage Firms, Broker Agents
Client Portal	Web + Mobile View	Direct Buyers, Existing Clients

E. TARGET USERS & STAKEHOLDERS

المستخدمون وأصحاب
المصلحة

USER TYPE	ROLE DESCRIPTION	PRIMARY PLATFORM
Direct New Client	First-time buyer discovering the developer	Mobile App / Client Portal
Existing Client	Client with active contract and payment plan	Mobile App / Client Portal
Broker Agent	Individual real estate agent with a brokerage	Broker Portal / Mobile App
Brokerage Firm Manager	Manager overseeing agents and commissions	Broker Portal
Sales Staff	Internal sales team managing leads and conversions	Admin Dashboard / Mobile
Finance Staff	Managing payments, invoices, collections	Admin Dashboard
Customer Service	Handling client inquiries and complaints	Admin Dashboard
Maintenance Technician	Resolving maintenance tickets	Mobile App
Maintenance Manager	Overseeing maintenance operations	Admin Dashboard
Project Manager	Managing construction and delivery timelines	Admin Dashboard

USER TYPE	ROLE DESCRIPTION	PRIMARY PLATFORM
Legal / Contracts Team	Managing and reviewing contracts	Admin Dashboard
Senior Management	Executive dashboards and reports	Admin Dashboard
System Administrator	Platform configuration, access control	Admin Dashboard
Security / Compliance	Monitoring fraud, audits, risk management	Admin Dashboard



F. FULL USER JOURNEYS

رحلة المستخدمين الكاملة

F.1

New Client Journey

1. Client discovers a new project via social media, referral, or broker
2. Downloads mobile app or accesses web portal
3. Registers account: name, phone, email, national ID, address
4. Uploads identity documents: national ID + passport (if applicable)
5. Completes facial recognition / biometric verification linked to uploaded ID
6. KYC check passes; account activated (or flagged for manual review)
7. Client browses project details: unit types, floor plans, pricing, payment plans
8. Client expresses interest: submits EOI form and pays EOI amount online
9. System assigns priority number in reservation queue based on time + payment confirmation
10. Client receives notification of scheduled appointment date and time to visit the office
11. Client attends office on scheduled date, selects preferred unit
12. Client has 7 days to sign the physical contract at the developer's office
13. Client books contract signing appointment via the app
14. Contract uploaded to platform; payment plan auto-extracted by AI
15. Client sees full payment dashboard: paid amounts, upcoming installments, overdue alerts
16. Client receives reminders before each installment due date
17. Client pays installments through integrated payment gateway
18. Platform tracks delivery timeline; client receives unit handover notification
19. Post-handover: client activates community services, submits maintenance requests, requests access cards

F.2 Broker / Brokerage Firm Journey

20. Brokerage firm contacts the developer; formal brokerage agreement signed
21. Firm registers on broker portal: company name, commercial license, tax documents, signed brokerage contract
22. Platform reviews and approves broker account
23. Broker creates a client request: enters client name, generates a secure personalized registration link
24. Client receives link, self-registers, uploads documents, and completes verification
25. Client is permanently linked to the broker in the system
26. Client pays EOI or reservation amount; broker's ownership claim is locked
27. Broker tracks client status, appointment schedule, and unit selection via the portal
28. Post-sale: broker tracks commission status — pending, approved, paid, rejected
29. Broker performance dashboard shows units sold, total sales value, and ranking

F.3 Existing Client Journey

30. Client logs in with biometric authentication
31. Views home dashboard: unit details, project, payment status, remaining balance
32. Checks upcoming installments and pays online
33. Tracks delivery status and construction progress
34. Post-handover: submits maintenance requests, uploads photos/videos
35. Tracks ticket status in real time; receives resolution confirmation
36. Requests community services: visitor access, parking, utility connections, access cards

G. PERMISSION MATRIX

مصفوفة الصلاحيات

ROLE	PERMISSION DESCRIPTION
Super Admin	Full system access. User management, configuration, all modules, audit logs, security controls.
Developer Manager	All operational views. Approve/reject users, contracts, commissions. Financial dashboards. Reports.
Sales Manager	Manage sales team. Assign leads. View all units and project inventory. Approve reservations.
Sales Staff	Register clients, initiate reservations, view unit availability, log communication, book appointments.
Finance Manager	Full payment dashboard. Approve commissions, refunds, payment adjustments. ERP integration.
Finance Staff	View payments, generate receipts, flag overdue accounts, process collections.
Customer Service	View client profiles, log interactions, raise tickets, escalate issues, send notifications.
Maintenance Manager	Assign and manage tickets, set priorities, view technician performance, approve completions.
Maintenance Technician	View assigned tickets, update status, upload completion photos, log work done.
Legal / Contracts	Upload/review contracts, view contract status, flag issues, generate legal reports.
Project Manager	Manage delivery milestones, update construction progress, communicate with clients.
Brokerage Firm Manager	Manage broker agents, view firm-wide performance, track commissions, register clients.
Broker Agent	Register clients via secure links, track client status, view own commissions and sales.
Direct Client (New)	Register, verify identity, browse projects, pay EOI, book appointment, choose unit.
Direct Client (Existing)	View contract, pay installments, track delivery, submit maintenance, access community services.
Senior Management	Executive dashboards, KPI reports, project performance, risk alerts, no transactional access.

ROLE	PERMISSION DESCRIPTION
Security / Compliance	Audit logs, fraud alerts, user activity monitoring, risk flags, compliance reports.
System Administrator	Platform configuration, role management, API integrations, system health monitoring.
Marketing Manager	Manage marketing campaigns, track lead attribution, manage integrations with Facebook/Google/TikTok Lead Ads.
Call Center Agent	Manage inbound/outbound calls, access CRM profiles during calls, view call logs and records.
Collection Agent	Access overdue queues, record Promise to Pay (PTP) agreements, initiate legal default escalations.
Warranty Manager	Configure warranty terms, validate claim warranty coverage, assign vendor repair orders.
Vendor Coordinator	Manage vendor records, review contractor SLA metrics, assign outsourced work orders.

H.1 Registration & Identity Verification

- Full user registration with: full name, phone number, email, national ID number, passport number (optional), address
- User type classification at registration: new client, existing client, broker, brokerage firm, admin, sales, finance, maintenance
- Document upload module: national ID card (front/back), passport, signed contracts, commercial registration, tax certificates
- Facial recognition / biometric liveness check linked to uploaded government ID
- KYC-equivalent verification workflow: document authenticity, OCR data extraction, liveness match
- Anti-duplication engine: phone, email, national ID, and biometric cross-checking
- Manual review queue for flagged registrations with admin approval workflow
- Verification status tracking: pending, verified, rejected, under review
- Secure re-verification flow for document updates or identity changes

H.2 Existing Client Portal

- Client home dashboard with: unit summary, project name, building, floor, unit number, type, area
- Financial summary: total price, paid amount, remaining balance, overdue installments
- Full payment plan viewer with installment schedule, due dates, and amounts
- Payment history log with receipts and status (paid, partial, late, cancelled, refunded)
- Delivery timeline tracker: expected date, current construction stage, actual delivery status
- Contract viewer: downloadable signed contract, addenda, handover documents
- Maintenance request portal integrated into client dashboard
- Push notifications for upcoming installments, overdue alerts, delivery updates
- Multi-unit management for clients owning more than one unit

H.3 Payment System

- Online payment for: EOI amounts, reservation deposits, installments, maintenance fees, service fees
- Integration with certified payment gateways (e.g., Stripe, Paytabs, Fawry, local banking APIs)
- Multi-currency support with configurable exchange rates
- Payment status engine: paid, pending, overdue, partial, cancelled, refunded
- Automated receipt generation and email/SMS delivery
- Failed payment retry logic with client notification and grace period management
- Finance dashboard: collections tracker, aging reports, overdue alerts, payment forecasting
- ERP/accounting system integration (SAP, Oracle, Microsoft Dynamics, Odoo)
- Audit trail for every payment transaction with timestamps and user attribution
- Refund management workflow with approval chain

H.4 New Project Launch & Reservation System

- Admin publishes new project: name, description, location, unit types, floor plans, pricing, payment options, launch date
- Multi-channel announcement: push notification, in-app, email, SMS, WhatsApp (where permitted)
- Client-facing project page: full details, gallery, 3D renders, location map, price range, payment plan simulator
- EOI payment integrated into project page — client pays to enter reservation queue
- Priority queue engine: assigns position based on payment timestamp and client eligibility rules
- Automated scheduling system: assigns each client an appointment date/time to visit the office and select a unit
- Appointment reminder notifications (3 days, 1 day, 2 hours before)
- Unit selection interface: clients see available units on launch day, choose their preference
- 7-day contract signing window with in-app appointment booking for physical office visit
- Waiting list management for units freed by cancellations or no-shows



H. COMPLETE FEATURE LIST (CONTINUED)

قائمة المميزات الكاملة —
تابع

H.5 Unit Inventory & Reservation Management

- Real-time unit inventory dashboard per project: available, reserved, sold, cancelled, pending, under negotiation
- Unit profile: unit number, building, floor, type, area (sqm), total price, status, linked client or broker
- Atomic reservation logic: prevents double-booking through database-level locking
- Permission-based unit allocation: only authorized users can reserve or sell units
- Status change audit trail: who changed the status, when, and from what previous state
- Automatic removal of reserved/sold units from available inventory view
- Bulk import and update of unit data via Excel/CSV for large projects
- Unit filtering and search: by type, area, price range, floor, status, linked entity
- Interactive floor plan view with color-coded availability

H.6 Broker & Brokerage Portal

- Brokerage firm registration with document upload: commercial license, tax certificate, signed brokerage agreement
- Firm-level and agent-level access separation: manager and staff roles within each brokerage
- Secure client link generation: unique, time-limited registration URLs tied to a specific broker
- Client registration via broker link: client self-registers, uploads ID, completes KYC
- Permanent broker-client link: once a client registers via broker link, they are associated permanently
- Anti-poaching system: client cannot be claimed by another broker once linked
- Conflict resolution rules: first verified registration wins; system flags disputes automatically
- Commission tracking: unit sold price, agreed commission rate, commission amount, payment status
- Commission status workflow: pending → approved → paid (or rejected with reason)
- Broker performance leaderboard: units sold, total value, conversion rate, client count
- Broker portal dashboard: client list, unit status, appointment tracker, commission overview

H.7 Administration Dashboard

- Real-time KPI dashboard: total registered users, verified/pending users, units available/sold/reserved
- Project management module: multiple concurrent projects, status tracking, delivery milestones
- Finance control panel: payment collections, overdue tracking, revenue forecasting, commission approval
- Sales performance: by agent, by team, by broker, by project
- Broker management: all registered brokers, performance metrics, commission pipeline
- Maintenance operations overview: open tickets, resolution times, technician assignments
- Customer service queue: open cases, escalations, response time SLAs
- User activity logs and audit trails: complete, tamper-proof records of all system actions
- Notification control center: create, schedule, and send announcements to any user segment
- Fraud and security alerts dashboard: real-time suspicious activity monitoring
- Role management: create, edit, and assign roles and permissions

H.8 Post-Handover & Maintenance Services

- Maintenance request submission: category, description, priority indication, photo/video upload
- AI-powered ticket categorization: automatically classifies by type and routes to the correct department
- Ticket lifecycle management: open, assigned, in-progress, resolved, confirmed, closed
- Escalation engine: auto-escalates tickets exceeding SLA response time
- Client satisfaction rating: post-resolution star rating and comment
- Community services module: visitor access permits, parking requests, move-in/move-out scheduling
- Utility connection requests: electricity, water, gas, internet
- Access card management: request, issue, replace, revoke
- Emergency request channel: priority routing for urgent safety issues
- Amenity booking system: gym, pool, meeting rooms, recreation areas
- Housekeeping and landscaping service requests
- Post-handover checklist: guided onboarding for new residents after receiving keys

✓ H. COMPLETE FEATURE LIST (CRM & MARKETING)

قائمة المميزات — إدارة علاقات العملاء والتسويق

H.9 Complete Sales CRM Pipeline

- **Full Deal Pipeline Workflow:** Visualizes and tracks lead progression through standard corporate stages: New Lead → Contacted → Interested → Visit Scheduled → Negotiation → Reservation → Contract → Closed Won / Closed Lost.
- **Multi-Channel Activity Logging:** Auto-logs and tracks customer communication history including inbound/outbound Calls, WhatsApp messages, Email logs, physical Meeting schedules, and structured follow-ups.
- **Automatic Routing & Lead Assignment Engine:** Dynamically routes new leads to sales staff based on project specialization, language, target location, availability, or round-robin logic.
- **Lead Source Attribution:** Links lead origin (broker referral, social campaign, direct walk-in, website form) directly to the CRM profile for transparent conversion metrics.
- **Deal Performance Metrics:** In-depth stats on deal closure speed, conversion rate per agent/team, and total sales pipeline value tracking.

H.10 Unified Call Center Module

- **Integrated VoIP Softphone:** Empowers agents to place and receive calls directly inside the web/mobile app dashboard with automatic system recognition.
- **Bi-directional Call Management:** Inbound queuing with priority routing based on client classification and outbound campaign auto-dialing.
- **Call Recording & Archiving:** Securely records and archives calls, embedding them directly into the client's CRM history card for quality control and audits.
- **Interactive Voice Response (IVR):** Custom IVR flow builder to route callers (e.g., "Press 1 for Sales, Press 2 for Billing, Press 3 for Maintenance").
- **Agent Productivity Metrics:** Real-time tracking of calls handled, average call duration, wait times, wrap-up times, and call resolution rates.

H.11 Advanced Marketing Automation Module

- **Multi-Channel Lead Sync:** Direct real-time API integrations with Facebook Lead Ads, Google Ads, and TikTok Lead Generation forms.
- **Granular UTM Tracking & Analytics:** Records `utm\_source`, `utm\_medium`, `utm\_campaign`, `utm\_content`, and `utm\_term` for all incoming web leads.
- **Customer Origin Mapping:** Instant reporting on exactly where each client came from, identifying the exact campaign, ad set, and creative that triggered the conversion.
- **Automated Nurturing Flows:** Trigger-based email and SMS nurturing sequences based on interest score, brochure downloads, or appointment cancellations.
- **ROI Analysis Engine:** Cross-references marketing spend with closed sales down-payments to compute live Customer Acquisition Cost (CAC) and Campaign ROI.

H.12 Dynamic Inventory Pricing Engine

- **Demand Monitoring System:** Continually tracks system indicators including unit views count, brochure downloads, and reservation queue velocity.
- **Algorithmic Price Recommendations:** Suggests automated price adjustments (upwards for hot projects/views, discounts for slow inventory) based on predefined thresholds.
- **Approval & Lock Workflow:** Suggested price revisions trigger a multi-level approval pipeline (Sales Manager → Developer Manager → CEO) before applying.
- **Dynamic Floor & View Premiums:** Real-time multipliers applied to base price based on view excellence (e.g., pool, sea, street) and floor elevation demand.
- **Competitor Pricing Intelligence:** Structured module for analysts to input competitor data to calculate price elasticity models.

H. COMPLETE FEATURE LIST (FINANCIAL OPERATIONS & CANCELLATIONS)

قائمة المميزات —
العمليات المالية والإلغاء

H.13 Installment Rescheduling Workflow

- **Client-Initiated Modifications:** Portal request interface allowing existing clients to request installment deferral or payment plan reshaping.
- **Rescheduling Impact Engine:** Calculates live cash flow impact, outstanding balance adjustments, and applies late interest/rescheduling fees.
- **Multi-Level Approval Pipeline:** Direct routing of proposed payment plans through the authorization chain: Customer Relations → Finance Manager → Executive Board.
- **Contract Addendum Automation:** Auto-generates legal reschedule addendums upon approval, queuing them for physical or validated digital signature.
- **Reschedule Audit Trail:** Keeps a history of payment plan changes, showing original vs. revised schedules and who authorized the modifications.

H.14 Dedicated Collections Module

- **Collections Team Dashboard:** Real-time queue displaying overdue accounts categorized by aging buckets (30, 60, 90, 120+ days late).
- **Promise to Pay (PTP) Management:** Records client commitments to settle outstanding dues (date, amount, installment target) with automated follow-ups.
- **Automated Dunning Sequences:** Trigger-based email, SMS, and registered mail sequences sent to delinquent accounts according to escalation phases.
- **Legal Escalation Workflow:** Automatically routes client files to the Legal Team module when default exceeds 90 days, flagging contractual breach protocols.
- **Collection Performance Analytics:** Measures key team metrics including Collection Efficiency Index (CEI), PTP adherence rate, and aging reduction.

H.15**Structured Cancellation Workflow**

- **Initiation Portal:** Handles client-requested cancellations or system-enforced defaults due to non-payment terms.
- **Deduction & Penalty Calculator:** Computes administrative fees, cancellation penalties, and tax deductions based on the contract's refund clauses.
- **Automated Revert to Inventory:** Immediately updates the unit status back to "Available" in the dynamic inventory, making it bookable.
- **Refund Settlement Queue:** Financial workflow managing installment refund payouts to the client, including multi-stage manager signoffs.
- **Cancellation Audit:** Tamper-proof history logging cancellation causes, total client contributions, refund payouts, and legal documentation.

H.16**Comprehensive Vendor & Contractor Management**

- **Contractor/Vendor Registry:** Directory of verified third-party contractors, suppliers, and service providers with active compliance records.
- **SLA Monitoring & Management:** Configurable service level agreements (SLAs) per vendor mapping expected response, resolution, and delivery times.
- **Outsourced Work Orders:** Generates and dispatches work orders directly to vendors for complex maintenance/handover snag resolution.
- **Performance Rating System:** Dynamic scoring based on client reviews, completion speed, and quality audits, affecting future assignments.
- **Integrated Billing & Payments:** Auto-reconciles vendor invoices against approved work completions, feeding directly into the general ledger.

H. COMPLETE FEATURE LIST

(HANDOVER, WARRANTY & SYSTEM CORE)

قائمة المميزات — التسليم
والضمان ومحرك النظام

H.17 Advanced Handover Management

- **Pre-Handover Inspection Interface:** Guided mobile walk-through app for quality engineers to inspect and record issues before delivery.
- **Defect Snagging & Photo Uploads:** Log defect locations, add notes, select severity, and attach high-res photo snags directly to the unit file.
- **Digital Engineer & Client Sign-off:** Secure digital capture of engineer validation and client signature upon walk-through acceptance.
- **Automated Handover Certificate:** Generates legally binding, client-signed Handover Certificates upon zero-snap inspection completion.
- **Defect Resolution Tracking:** Direct routing of snag items to internal maintenance or external vendors with countdown timers against SLAs.

H.18 Unit Warranty Management System

- **Warranty Categories Mapping:** Multi-tier warranty scheduling based on components (e.g., Finishing: 6M, Plumbing: 1Y, Electrical: 2Y, Structural: 10Y).
- **Auto-Validity Checks:** Automatically checks warranty eligibility when a resident submits a post-handover maintenance ticket.
- **Claim Settlement Workflow:** Manages warranty claim approvals, routing free repair orders to original subcontractors or system vendors.
- **Warranty Expiry Notifications:** Proactively notifies clients before their component warranty periods expire, offering optional extension packages.
- **Warranty Cost Center:** Tracks developer expenditure on warranty repairs, helping identify high-failure equipment or poor-quality subcontractors.

H.19 Configurable Visual Workflow Engine

- **No-Code Workflow Builder:** Visual drag-and-drop rule engine for administrators to construct trigger-action sequences without redeploying code.
- **Event Trigger Setup:** Triggers sequences based on events (e.g., "Installment Overdue", "KYC Verified", "Unit Reserved", "Snag Logged").
- **Conditional Branching Logic:** Sets path rules (e.g., "If Overdue > 30 Days → Send SMS; If Overdue > 60 Days → Call; If Overdue > 90 Days → Collections").
- **Multi-Channel Dispatch Action:** Sends automated push notices, SMS notifications, email alerts, WhatsApp business messages, or logs manual CRM tasks.
- **Execution Monitor:** Dashboards displaying workflow success rates, queue status, active templates, and detailed execution logs.

H.20 Document Management System (DMS) with OCR

- **OCR-Powered Full-Text Search:** Scans, indexes, and enables keyword search within PDFs, scanned national IDs, contracts, and commercial licenses.
- **Document Version Control:** Maintained log of all draft versions, updates, and final signoffs for contracts and legally binding addendums.
- **Automatic Expiry Notifications:** Auto-extracts expiry dates from commercial licenses, broker agreements, and IDs, alerting owners 30 days prior.
- **Corporate Retention Policies:** Configurable rules dictating document storage lifecycles, archival timelines, and permanent purge policies.
- **Granular Document Access:** Enforces document-level permissions, preventing unauthorized download, editing, or viewing of PII and contracts.

H.21 Corporate Executive BI Dashboard

- **Revenue Forecasting:** Predictive cash inflows based on active payment plans, collection efficiency trends, and projected project launches.
- **Live Cash Flow Forecasts:** Compares active collection schedules against project outflow budgets and scheduled constructor payouts.
- **Sales Velocity Tracking:** Measures monthly sales speed per project, unit type, and marketing campaign.
- **Unsold Inventory Valuations:** Dynamic reports illustrating the total book value of remaining unsold units with premium adjustments.
- **Collection & Broker Efficiency Index:** Displays live performance scorecards for collection agents and external brokerage organizations.



I. MOBILE APP FEATURES (IOS & ANDROID)

مميزات تطبيق
الهواتف

Client Features

FLUTTER

- Biometric login: Face ID, fingerprint, or PIN
- Home dashboard: unit summary, payment status, upcoming installments, delivery progress
- Project browser: new launches, unit availability, floor plans, price calculator
- EOI and reservation payment in-app
- Appointment scheduling and management
- Contract viewer and document downloads
- Maintenance request with camera integration
- Community services: access cards, visitor permits, utility requests
- Push notifications for all critical events
- In-app chat with customer service team
- AI-powered chatbot for instant query resolution

Broker & Staff Features

FLUTTER

- Client management: add clients, generate registration links, track status
- Commission tracker: real-time commission pipeline and payment status
- Appointment calendar for client office visits
- Performance summary: units sold, revenue generated, ranking
- Sales staff: lead management, appointment booking, unit search
- Maintenance technicians: view assigned tickets, update status, upload photos
- Field access to client data within permission scope



J. WEB APPLICATION / DESKTOP FEATURES

مميزات تطبيق
الويب

- All mobile features plus advanced data tables, bulk operations, and complex reports
- Advanced unit inventory management with floor plan overlay editor
- Full contract management: upload, review, annotate, version control
- Bulk client import via CSV or Excel for legacy data migration
- Advanced payment reports: custom date ranges, export to Excel/PDF

- ERP and accounting system integration interface
- Advanced broker management: mass commission approvals, bulk communication
- Full audit trail viewer with advanced filtering and export
- Document management: organized vault for all contracts, IDs, and legal documents
- Advanced analytics and BI dashboards with drill-down capabilities
- Multi-project comparison views for management decision-making
- System configuration interface for admins

N. PAYMENT SYSTEM — TECHNICAL SPECIFICATION

المواصفات الفنية لنظام
الدفع

PAYMENT TYPE	TRIGGER	NOTES
EOI (Expression of Interest)	Project launch phase	Refundable if unit not allocated
Reservation Deposit	After unit selection	Locks the unit for 7 days
Contract Down Payment	Contract signing	Non-refundable, part of total price
Installment Payment	Per payment plan schedule	Recurring; automated reminders
Maintenance Fee	Annual or as needed	Compound operational cost
Service Fee	On-demand	Utility connection, access card, etc.
Penalty / Late Fee	Overdue installments	Per contract terms
Refund Processing	Cancellation or error	Admin-approved workflow

CRITICAL Payment Gateway Integration

- Primary: region-appropriate payment gateway (e.g., Paytabs, Fawry, Stripe)
- Tokenized card storage for returning clients (PCI-DSS compliant)
- 3D Secure authentication for card payments
- Bank transfer support with auto-reconciliation
- Digital wallet support (Apple Pay, Google Pay)
- Payment failure handling: retry logic, grace period notification, account hold

O. CONTRACT MANAGEMENT SYSTEM

نظام إدارة العقود

- Contract upload: PDF or scanned document upload by admin or legal team
- AI-powered contract data extraction: client name, unit details, total price, payment plan, handover date
- Payment plan auto-import: AI reads uploaded payment schedule tables and populates the system automatically
- Contract version control: track amendments, addenda, and updates
- Contract status workflow: draft → under review → signed → active → completed → cancelled

- Contract signing appointment: client books a time slot via app for physical signature at office
- 7-day contract signing window enforced by system with automated reminders
- Contract viewer: clients can view their signed contract in the portal at any time
- Contract expiry alerts: notifications for unsigned contracts approaching the deadline
- Legal team review workflow: flag, comment, and approve/reject contract submissions
- Handover documentation: pre-delivery inspection forms, snag list, acceptance certificates
- Document vault: secure, organized storage of all contracts per client and unit



R. FRAUD PREVENTION & COMPLIANCE

نظام منع الاحتيال
والامتثال

CRITICAL

Identity Fraud Prevention

- Biometric liveness detection during registration: prevents photo substitution
- OCR extraction and cross-validation of national ID and passport data
- AI-powered document authenticity check: detects tampering or forgery
- Duplicate identity detection: same national ID, phone, email, or biometric profile
- Multi-account detection: one person attempting multiple registrations

CRITICAL

Transaction Fraud Prevention

- Anomaly detection for unusual payment patterns or amounts
- Unauthorized unit allocation alerts
- Double reservation prevention: database-level locking
- Commission fraud detection: false claims or inflated prices
- Rapid reservation pattern alert

HIGH

Internal Misconduct Prevention

- All staff actions logged with timestamp, user ID, and IP address
- Unauthorized access alerts: staff accessing data outside permission scope
- Approval workflows: no high-value action without multi-level approval
- Audit trail immutability: logs cannot be edited or deleted by any user

HIGH

Broker Conflict Management

- Client ownership lock: first broker to register and verify a client owns that lead permanently
- Automatic dispute detection: same client registered under two brokers triggers alert
- Manual dispute resolution workflow with documented outcome
- Broker agreement enforcement: only active approved brokers can operate



S. NOTIFICATION SYSTEM

نظام الإشعارات

NOTIFICATION TYPE	CHANNELS	RECIPIENT
Installment Due Reminder	Push, SMS, Email, WhatsApp	Client
Overdue Payment Alert	Push, SMS, Email	Client + Finance Team
New Project Announcement	Push, Email, SMS, WhatsApp	All Eligible Clients + Brokers
Reservation Queue Update	Push, In-app	Client
Appointment Confirmation	Push, Email, SMS	Client
Contract Signing Reminder	Push, SMS, Email	Client + Legal Team
Delivery Update	Push, Email	Client
Maintenance Ticket Update	Push, In-app	Client + Technician
Commission Status Update	Push, Email	Broker
Fraud Alert	In-app, Email	Security + Management
Emergency Maintenance	Push, SMS	Maintenance Manager



Q. ARTIFICIAL INTELLIGENCE FEATURES

مميزات الذكاء
الاصطناعي

AI

Document Intelligence

- OCR and NLP contract parsing: auto-extract client name, unit, price, payment plan, handover date
- Payment schedule extraction: reads uploaded installment tables and auto-populates system
- Identity document processing: extracts and validates data from national ID and passport
- Document authenticity scoring: flags potentially tampered or fraudulent documents

AI

Client, Broker & CRM Sales Intelligence

- AI chatbot (Arabic + English): answers client queries 24/7
- Broker chatbot: provides instant answers on commission rules, unit availability
- Lead scoring: predicts likelihood to purchase
- Project recommendations based on client profile and preferences
- Communication summarization for quick staff briefing
- **AI Sales Assistant:** Predicts closing probability, recommends optimal unit selections, and calculates the best time to call/contact leads based on history.

AI Operational & Financial Forecasting

- Maintenance ticket classification: auto-categorizes and routes to correct team
- Priority scoring for maintenance: identifies urgent vs. routine automatically
- Customer satisfaction NLP analysis of feedback and reviews
- Payment default prediction based on payment history patterns
- Broker performance analytics: AI-powered insights on productivity
- **AI Cash Flow & Sales Forecasting:** Predicts liquidity requirements, project financial margins, delinquency risks, and next-month sales volumes.
- **AI Pricing Suggestions:** Recommends unit base price optimizations in real-time according to page views, reservation frequency, and remaining project capacity.

AI Security Intelligence

- Anomaly detection: unusual patterns in user behavior or access times
- Duplicate client detection via biometric data and behavioral signals
- Risk scoring: assigns a risk score to every registered user
- Fraud pattern recognition: learns from historical cases



T. SECURITY REQUIREMENTS & COMPLIANCE

متطلبات الأمان
والامتثال

Authentication & Data Protection

- MFA enforced for all staff, admin, and broker accounts
- Biometric authentication: Face ID and fingerprint for mobile
- Session management: auto-logout, concurrent session prevention
- RBAC: every feature controlled by user role
- End-to-end encryption (TLS 1.3), AES-256 at rest
- PII data masking in all logs

Infrastructure & Backup

- PCI-DSS Level 1 compliance for payments
- Cloud deployment with SOC 2 Type II certification
- WAF and DDoS protection
- Automated daily backups (30-day retention)
- RTO < 4 hours, RPO < 1 hour
- Quarterly disaster recovery drills

U. DATABASE ARCHITECTURE OVERVIEW

نظرة عامة على معمارية قاعدة البيانات

ENTITY	KEY ATTRIBUTES	RELATIONSHIPS
users	user_id, type, name, phone, email, national_id, status, verified_at	→ Contracts, Payments, Tickets, Appointments
projects	project_id, name, location, launch_date, status, total_units	→ Units, Announcements, Reservations
units	unit_id, project_id, building, floor, unit_no, type, area, price, status	→ Reservations, Contracts, Clients
reservations	reservation_id, unit_id, user_id, broker_id, status, timestamp, priority_no	→ Units, Users, Brokers
contracts	contract_id, unit_id, client_id, signed_date, total_price, status	→ Payment Plans, Documents
payment_plans	plan_id, contract_id, installments[], due_dates[]	→ Payments, Contracts
payments	payment_id, plan_id, amount, paid_date, method, status, receipt_url	→ Payment Plans, Users
brokers	broker_id, firm_id, agent_name, status, commission_rate	→ Clients, Commissions, Contracts
commissions	commission_id, broker_id, unit_id, amount, status, approved_by	→ Brokers, Units, Payments
maintenance_tickets	ticket_id, unit_id, client_id, category, priority, status, assigned_to	→ Units, Users, Technicians
appointments	appointment_id, user_id, type, datetime, status, notes	→ Users, Units, Projects
audit_logs	log_id, user_id, action, entity, entity_id, timestamp, ip_address	→ All Entities
documents	doc_id, user_id, type, storage_url, verified, uploaded_at, version, expiry_date	→ Users, Contracts
notifications	notif_id, user_id, channel, message, sent_at, read_at	→ Users
leads	lead_id, name, phone, email, source, status, assigned_to_agent_id, ai_score,	→ Users, Interactions

ENTITY	KEY ATTRIBUTES	RELATIONSHIPS
	created_at	
interactions	interaction_id, lead_id, type (call, WhatsApp, email, meeting), notes, agent_id, timestamp	→ Leads
call_logs	call_id, direction (inbound/outbound), recording_url, duration, agent_id, status, linked_lead_id, client_id	→ Leads, Users
campaigns	campaign_id, name, channel, budget, leads_generated, conversions, ROI, utm_tags	→ Leads
rescheduling_requests	request_id, contract_id, proposed_installments[], reason, fee_applied, status, approved_by	→ Contracts
collections_queue	collection_id, contract_id, overdue_amount, days_overdue, calls_made, ptp_date, ptp_amount, status	→ Contracts
cancellations	cancellation_id, contract_id, penalty_deducted, refund_amount, revert_status, refund_status	→ Contracts
defects_snags	defect_id, unit_id, category, description, photo_url, severity, assigned_engineer_id, client_signoff, status	→ Units, Users
warranties	warranty_id, unit_id, component, period_months, start_date, end_date, vendor_id, status	→ Units, Vendors
vendors	vendor_id, name, category, SLA_rating, contact_info, active_contracts_count	→ Warranties, Defects
workflow_templates	template_id, name, trigger_event, conditions, actions_json, status	→ Audit Logs

W. REQUIRED INTEGRATIONS

التكامل المطلوب مع الأنظمة الخارجية

INTEGRATION	PURPOSE	PRIORITY
Payment Gateway	Online payment processing	CRITICAL — PHASE 1
SMS Gateway (Twilio/SNS)	OTP verification, payment reminders	CRITICAL — PHASE 1
Email (SendGrid/SES)	Transactional emails, notifications	CRITICAL — PHASE 1
WhatsApp Business API	Client notifications	HIGH — PHASE 2
Push (Firebase/APNs)	Mobile app push notifications	CRITICAL — PHASE 1
OCR/Document AI	Contract and ID data extraction	HIGH — PHASE 1
Facial Recognition API	Biometric KYC verification	HIGH — PHASE 1
ERP / Accounting System	Financial reconciliation	HIGH — PHASE 2
Google Maps / Mapbox	Project location display	MEDIUM — PHASE 1
OpenAI / Azure OpenAI	Chatbot and NLP features	HIGH — PHASE 2

Z. IMPLEMENTATION PHASES

مراحل التنفيذ

PHASE 1 MVP — Months 1–6

Goal: Deliver a fully functional core platform covering registration, inventory, basic payments, and client portal.

- User registration and KYC verification (document upload, facial recognition)
- Role-based access control and user management
- Unit inventory management: real-time availability, status tracking
- Basic payment processing: installments, EOI, reservation deposits
- Client portal: contract view, payment plan, dashboard
- Broker portal: client registration links, commission tracking basics
- Admin dashboard: user management, unit management, financial overview
- Notification system: push, email, SMS
- New project launch and priority queue system
- Mobile app: iOS and Android (core features)
- Payment gateway integration

PHASE 2 Enhanced Platform — Months 7–12

- AI chatbot for clients and brokers (Arabic + English)
- AI document processing: contract data extraction and payment plan import
- Advanced broker management: performance leaderboard, commission workflows
- Full post-handover maintenance and community services module
- ERP/accounting system integration
- Advanced fraud detection system
- Interactive floor plan viewer with unit selection
- Advanced analytics and BI reporting dashboards

PHASE 3 Advanced & Innovation — Months 13–18

- AI-powered payment default prediction and proactive collection management
- Virtual property tours: 360-degree views and VR integration
- Digital contract signing (e-signature with legal validity)
- Predictive maintenance: AI analysis of recurring issues
- Smart community features: IoT integration for access control, energy management
- Multi-developer platform: license to other real estate companies

PHASE 4 Enterprise REOS Expansion — Months 19–24

- **Full CRM & Communications Integration:** Full pipeline automation, VoIP call center softphones, activity logs, and auto-routing rules.
- **Marketing & Campaign Automation:** Lead Ads direct integrations, granular UTM metrics, customer origin reports, and nurturing flows.
- **Dynamic Pricing Suggestions Engine:** Real-time demand trackers, competitor pricing input, and automatic price recommendation algorithms.
- **Installment Rescheduling & Collections Module:** Rescheduling calculation calculators, dunning workflows, Promise to Pay (PTP) logs, and legal default escalations.
- **Snag Inspection & Warranty Control:** Pre-handover snag inspection app, digital walk-through signature captures, multi-tier warranty claims, and vendor SLA control panels.
- **Configurable Workflow Engine:** No-code trigger-action visual flow mapping for notification schedules and team escalations.
- **Document Management System (DMS):** OCR-powered text scanning, draft version history, expiry alert flows, and retention schedules.
- **Executive BI Dashboard:** Financial cash flow forecasts, revenue forecasting, unsold inventory valuation maps, and broker efficiency indexing.



BB. RISK REGISTER & CC. SUCCESS KPIs

سجل المخاطر ومؤشرات الأداء

RISK	PROBABILITY	MITIGATION STRATEGY
Scope creep extending timeline	HIGH	Strict MVP scope control; phased delivery; change request process
Low user adoption on launch	MEDIUM	Change management program; staff training; client communication campaign
Integration delays (ERP, payment)	MEDIUM	Early vendor engagement; API documentation review; parallel development
Data migration from legacy systems	HIGH	Dedicated migration sprint; data validation; parallel running period
Security breach or data leak	LOW	Security-first architecture; penetration testing; incident response plan
Performance under 100K+ users	MEDIUM	Load testing before launch; auto-scaling cloud infrastructure
Payment gateway failures	LOW	Multi-gateway setup; automatic failover; payment retry logic

KPI	TARGET	MEASUREMENT
User registration completion rate	> 85%	Registered / Started registration
KYC verification success rate	> 90%	Verified / Submitted
Payment collection rate	> 95%	Paid on time / Total due
Maintenance ticket resolution time	< 48 hours avg	Resolution date - Open date
Client satisfaction score	> 4.2/5.0	Post-resolution survey
Unit reservation accuracy	0 double bookings	Conflict incidents per month
Fraud detection rate	> 95% true positive	Flagged vs. confirmed fraud

KPI	TARGET	MEASUREMENT
System uptime	> 99.9%	Monthly availability monitoring

DD. DECOUPLED DEVELOPMENT TASK

DIVISION

تقسيم كامل المهام البرمجية المستقلة
بالتفصيل

لضمان أقصى درجات الاستقلالية وعدم الاعتمادية (Zero Dependency) وتجنب أي تعطيل برمّي أثناء العمل بالتوازي (In Parallel)، تم تقسيم **جميع مميزات النظام الـ 21 (H.1 إلى H.21) بالإضافة للواجهات والمواصفات ونظم منع الاحتيال** بشكل كامل ورأسي (Vertical Slicing) على المطورين الثلاثة. يتولى كل مطور برمجة الجزء الخاص به بالكامل من جداول قاعدة البيانات والـ API وواجهات الويب والموبايل والربط الخارجي:

المطور المسئول	الموديولات والوظائف (MODULES & FEATURES)	جداول قاعدة البيانات (DATABASE ENTITIES)
 رجب (Ragab) ACQUISITION & SALES	→ H.1 التسجيل والتحقق من الهوية (KYC) → H.4 إطلاق المشاريع ونظام الحجوزات → H.6 بوابة ومبيعات السماسرة (Brokers) → H.9 نظام الـ CRM ومراحل المبيعات → H.10 سنترال الـ VoIP وتسجيل المكالمات → H.11 ربط التسويق المباشر (FB/TikTok) → H.7 لوحة أداء المبيعات والسماسرة → تطبيقات السماسرة ومبيعات الموبايل (I) → واجهات الـ CRM والاتصالات الويب (J) → منع احتيال الهويات والتسجيل (R.1)	,leads ,interactions ,call_logs ,campaigns ,brokers commissions
 ملواني (Melwany) CONTRACTS & FINANCE	→ H.3 نظام وبوابات الدفع الإلكتروني → H.5 جرد المخازن وحجز الوحدات الديناميكي → H.12 محرك التسعير الديناميكي للوحدات → H.13 طلبات جدولة وتأجيل الأقساط → H.14 تحصيل الديون المتأخرة والـ PTP	,units ,projects ,reservations ,contracts ,payment_plans ,payments ,collections_queue cancellations

المطور المسنول	الموديولات والوظائف (MODULES & FEATURES)	جداول قاعدة البيانات (DATABASE ENTITIES)
	→ H.15 إجراءات إلغاء التعاقد والغرامات → H.7 لوحات التحصيل والمالية والعمولات → تطبيقات دفع وحسابات الملاك الموبايل (I) → بوابات الدفع وجرّد المخازن الويب (J) → N + O العقود والدفع الفني والـ ERP → منع احتيال المعاملات والحجوزات (R.2)	
 محمود (Mahmoud) DELIVERY & INFRASTRUCTURE	→ H.2 بوابة العملاء الحاليين ولوحة التحكم → H.8 طلبات الصيانة وخدمات المجتمع → H.16 إدارة المقاولين والموردين (SLAs) → H.17 الفحص الفني للوحدات وتصوير العيوب → H.18 نظام إدارة فترات الضمان للبناء → H.19 محرك العمليات البصري (Workflow) → H.20 الأرشفة الذكية OCR و H.21 لوحة الـ BI → H.7 لوحة التحكم وصلاحيات المديرين وسجلات النظام → تطبيقات صيانة الملاك ومهندس الجودة الموبايل (I) → واجهات الـ OCR والـ BI والتحكم الويب (J) → S نظام الإشعارات العام (Push/SMS/WhatsApp) → منع التلاعب الداخلي وحسابات المراقبة (R.3)	,defects_snags ,warranties ,vendors ,workflow_templates ,documents ,maintenance_tickets appointments

DD. DECOUPLED DEVELOPMENT TASK DIVISION (CONTINUED)

تفاصيل عمل كل مطور
واستقلاليته

1 خطة عمل المهندس: رجب (Ragab) — محرك المبيعات والتسويق وجلب العملاء (Acquisition Engine)

الهدف الأساسي: جذب العملاء المهتمين والسماسة الجدد وتسجيل بياناتهم والتحقق من هوياتهم بشكل آلي وتوجيههم لخط سير المبيعات، ومراقبة أداء المكالمات والحملات.

- **تسجيل بيانات الهوية وال (H.1) KYC:** برمجة رفع الصور الشخصية والهوية، التحقق البصري ومطابقة ملامح الوجه بالفيديو ($Liveness\ Face\ Match > 85\%$)، واستخراج البيانات آلياً بال OCR، ومنع تكرار الحسابات الحيوية.
- **الحجوزات الأولى وإطلاق المشاريع (H.4):** برمجة بوابة دفع حجز EOI (جدية الحجز)، وأولوية طابور الحجز بدقة الملي ثانية (Redis Sorted Sets)، وجدولة المواعيد الآلية للزيارة واختيار الوحدات.
- **بوابة السماسة والشركات العقارية (H.6):** بوابة تسجيل الوكلاء والشركات، وتوليد روابط الإحالة الفريدة والمحددة زمنياً للعملاء لمنع نزاعات السماسة (Lead Lock & Anti-poaching)، وأداة فض النزاعات الذاتية للعملاء المشتركين.
- **ال CRM المتكامل ومراحل المبيعات (H.9):** لوحة Kanban التفاعلية لمراحل المبيعات السبعة، ونظام التوجيه التلقائي الموزع (Round-Robin Lead Routing) بناءً على اللغة والتخصص العقاري.
- **سنترال ال VoIP الهاتفي المتكامل (H.10):** هاتف برمجي WebRTC مدمج داخل شاشات المبيعات، نافذة منبثقة لبيانات المتصل (Screen Pop)، وتسجيل المكالمات وأرشفتها تلقائياً بصيغة MP3 مشفرة في AWS S3.
- **الربط التسويقي وقنوات الدعاية (H.11):** سحب فوري للبيانات عبر Webhooks مع Facebook/Google/TikTok Lead Ads، وتتبع معلمات ال UTM وحساب live لتكلفة الاستحواذ على العميل (CAC) والعائد التسويقي (ROI).
- **الواجهات والأنظمة الموكلة إليه (Specs I/J & W):** تطبيقات السماسة والمبيعات على الموبايل، شاشات ال CRM والمكالمات الويب، والربط مع بوابات التحقق والسنترال.
- **جولات العقارات الافتراضية وثلاثية الأبعاد (Z.Phase 3):** تطوير منصة محاكاة جولات العقارات الافتراضية بنطاق 360 درجة وتقنيات ال VR لتمكين العملاء الدوليين من زيارة وفحص العقار أونلاين.

جداول قاعدة البيانات الخاصة به: leads, interactions, call_logs, campaigns, brokers, commissions

الهدف الأساسي: معالجة المعاملات المالية بشكل فائق السرعة، ومراقبة المخزون الفعلي، وصياغة العقود وتعديل خطط السداد وإلغائها تلقائياً دون أي تدخل مع أنظمة الدعاية والتسليم.

- **بوابات الدفع الإلكتروني وتكامل الـ (H.3 / N / O) ERP:** ربط بوابات السداد (Stripe/Fawry/Paytabs) لإصدار الفواتير والوصولات التلقائية، وتسجيل القيود المحاسبية (Journal Entries) آلياً في نظام المحاسبة المركزي (SAP/Odoo).
- **إدارة المخزون الفعلي والأقفال الصارمة (H.5):** برمجة نظام الأقفال البرمجية لقواعد البيانات (Database PostgreSQL) في Lock: `SELECT ... FOR UPDATE` لمنع حجز الوحدة لأكثر من عميل في نفس المليون ثانية.
- **محرك التسعير الديناميكي للوحدات (H.12):** خوارزمية ذكية تزيد أسعار الوحدات آلياً بناءً على تكرار الزيارة (Views) ونسب الحجوزات المقفلة، مع تدفق الموافقات الإدارية البرمجية للمديرين.
- **جدولة الأقساط وتأجيل الدفعات (H.13):** حاسبة حساب فوائد الجدولة وغرامات التأخير وإعادة توزيعها على الدفعات المستقبلية، وتوليد ملاحق العقود المالية تلقائياً بصيغة PDF.
- **طابور تحصيل الديون وعود السداد (H.14):** جدولة أعمار الديون (Aging Buckets)، طوابير عمل المحصلين الآلية، وتسجيل وعود السداد (Promise to Pay - PTP) ومتابعتها للتصعيد القانوني.
- **نظام الإلغاء وتصفية الحسابات (H.15):** حساب غرامات إلغاء العقد والمبالغ المستردة للعميل، وإخلاء الوحدة التلقائي وإعادتها فوراً للمخزون المتاح للبيع.
- **الواجهات والأنظمة الموكلة إليه (Specs I/J & R.2):** شاشات حسابات وأقساط الملاك على الموبايل، لوحات تحكم الإدارة المالية وجدول المخزون والأسعار على الويب، وحماية المعاملات من الاحتيال المالي والحجز المزدوج.
- **التوقيع الإلكتروني الذكي والتعاقد الرقمي (Z.Phase 3):** دمج تقنيات التوقيع الإلكتروني المعتمدة قانونياً (Legally Binding e-Signatures) وتصدير مستندات العقود الرقمية الآمنة مباشرة.

جداول قاعدة البيانات الخاصة به: units, projects, reservations, contracts, payment_plans, payments, collections_queue, cancellations

- الهدف الأساسي: تسليم الوحدات المباعة، وتوفير تجربة تشغيل مثالية للملاك، وتنسيق أعمال المقاولين، وتطوير الأدوات البرمجية الذكية لخدمة الإدارة العليا وصناع القرار.
- بوابة العملاء الحاليين والصيانة (H.2 / H.8): شاشات للملاك لمتابعة البناء، بلاغات الصيانة والشكاوى الفورية، حجز مرافق الكومباوند، وتصاريح دخول الزوار البوابة (QR Code).
- الفحص الفني للاستلام وتسجيل الملاحظات (H.17): تطبيق معاينة الجودة لمهندس التسليم (Snagging App)، تصوير العيوب وتصنيف خطورتها، والتوقيع الرقمي للعميل على محضر الاستلام.
- إدارة المقاولين والضمانات (H.16 / H.18): توجيه تذاكر الصيانة تلقائياً للمقاولين وتتبع زمن الاستجابة (SLAs)، التحقق التلقائي لضمانات البند والتشطيبات لكل تذكرة صيانة لمنع هدر الميزانية.
- محرك أتمتة العمليات البصري (H.19): واجهة سحب وإفلات (No-Code Rule Builder) لتصميم مسارات العمل التلقائية للتنبيهات والمهام بناءً على أحداث النظام.
- الأرشفة الذكية (H.20) OCR DMS: مسح وتصنيف العقود والوثائق وتوليد نصوص قابلة للبحث الكامل (Full-Text Search OCR)، وتنبيهات انتهاء التراخيص والهويات المرفوعة.
- لوحات تحليل ذكاء الأعمال (H.21): التنبؤ بالتدفق النقدي والسيولة (Cash Flow Forecasting) بالذكاء الاصطناعي بناءً على سلوك السداد، وتقارير أداء المقاولين والمبيعات.
- لوحة الإعدادات العامة والصلاحيات (H.7 / S): إدارة الصلاحيات التامة (RBAC)، سجلات المراقبة الأمنية (Audit Logs)، ونظام الإشعارات الموحد (FCM/SMS/WhatsApp Business API).
- الترخيص المتعدد وأنظمة الكومباوند الـ IoT (Z.Phase 3): ربط البوابات الذكية وتصاريح دخول الكومباوند بأنظمة تشغيل الـ IoT، وبناء معمارية الـ SaaS لتعدد الشركات العقارية المستأجرة (Multi-tenant Partitioning).

جداول قاعدة البيانات الخاصة به: users, defects_snags, warranties, vendors, workflow_templates, documents, maintenance_tickets, appointments, notifications, audit_logs

بروتوكول التشغيل الخالي من الاعتمادية

CORE DECOUPLING PROTOCOL

التواصل عبر الأحداث (Event-Driven Communication): يمنع الكود الصلب والتداخل. عندما يسجل رجب حجراً ناجحاً، يُطلق الكود حدثاً عاماً `reservation.confirmed` يحتوي فقط على المعرف الفريد للوحدة والعميل. يستمع كود ملواني المالي لهذا الحدث آلياً لإنشاء العقد وجدول الأقساط، ويستمع كود محمود للتسليم لفتح سجل فحص جديد للوحدة وجدولة تسليم المفاتيح. لا يحتاج أي مهندس للوصول لكود المهندس الآخر لتشغيل هذا التدفق.

 EE. DEVELOPMENT TEAM + FF. TESTING
+ GG. LAUNCH

فريق التطوير والاختبار
والإطلاق

ROLE	COUNT	RESPONSIBILITIES
Product Manager	1	Requirements, roadmap, stakeholder management, sprint planning
Business Analyst	1	Process mapping, user stories, acceptance criteria, testing
UI/UX Designer	2	Mobile and web design, user research, design system
Backend Dev (Senior)	2	Architecture, APIs, database design, security
Backend Dev (Mid)	2	Feature development, integration, unit testing
Frontend Dev (Next.js)	2	Web app, admin dashboard, portals
Mobile Dev (Flutter)	2	iOS and Android apps
AI/ML Engineer	1	Document AI, chatbot, fraud detection, recommendations
DevOps / Cloud Engineer	1	Infrastructure, CI/CD, monitoring, security
Security Engineer	1	Penetration testing, compliance, audit systems
QA Engineer	2	Test planning, functional, regression, performance testing
DBA	1	Schema design, optimization, backup, recovery

TESTING PHASE	TYPE	FOCUS AREAS
Phase 1	Unit Testing	Individual functions: payment calculation, KYC verification, permission checks
Phase 2	Integration Testing	API connections, payment gateway, notification delivery, ERP sync
Phase 3	UAT	All user journeys with real stakeholders: clients, brokers, staff
Phase 4	Load Testing	100,000 simulated users; peak reservation events; payment stress
Phase 5	Security / Pen Testing	OWASP Top 10; API security; injection attacks
Phase 6	Regression Testing	After every sprint; automated regression suite

TESTING PHASE	TYPE	FOCUS AREAS
Phase 7	Beta Testing	Controlled release to 500 real users from each group

HH. Final Recommendations

The success of this platform depends on disciplined, phased delivery with continuous stakeholder alignment. Below are the critical success factors:

- ♦ Start with a focused MVP — do not build all features simultaneously
- ♦ Security is non-negotiable from Day 1
- ♦ Plan for data migration from legacy systems early
- ♦ Build a change management program for internal staff
- ♦ Plan for scale: architect for 1,000,000+ users eventually
- ♦ Invest heavily in UX research with real clients and brokers
- ♦ Choose an experienced real estate technology vendor
- ♦ Broker onboarding is critical — their portal must be polished
- ♦ Measure everything from launch day — set up analytics on Day 1
- ♦ Establish clear governance: who owns the platform, who decides

REDP — REAL ESTATE DIGITAL PLATFORM

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